SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Manual Test Case Generation and Bug Reporting System for Online Hotel Booking System

CSA3717 – Software Testing

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INTRODUCTION

- **Growth of online booking:** Customers search rooms, book, pay and cancel online, so even a small defect can cause lost bookings and revenue.
- **Need for testing:** Errors in search, availability, payment or cancellation reduce user trust, so every function must be verified before release.
- **Manual testing:** Testers use the website like a real guest and check each feature by hand to find functional, usability and display problems.
- **Test case generation:** Test cases are designed step by step using Equivalence Partitioning, Boundary Value Analysis and Decision Tables.
- **Bug reporting:** Every defect is recorded with steps to reproduce, expected and actual result, severity and priority, then tracked until it is closed.

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OBJECTIVES

- **Requirement Analysis:** Study the hotel booking requirements such as login, search, booking, payment and cancellation.
- **Test Case Design:** Create clear and complete test cases using standard design techniques and a fixed template.
- **Test Execution:** Run the test cases manually and compare actual results with expected results.
- **Bug Reporting:** Report each failed case with steps, severity, priority and screenshots so developers can fix it quickly.
- **Defect Tracking & Analysis:** Track defects from New to Closed, retest the fixes and prepare a test summary report.

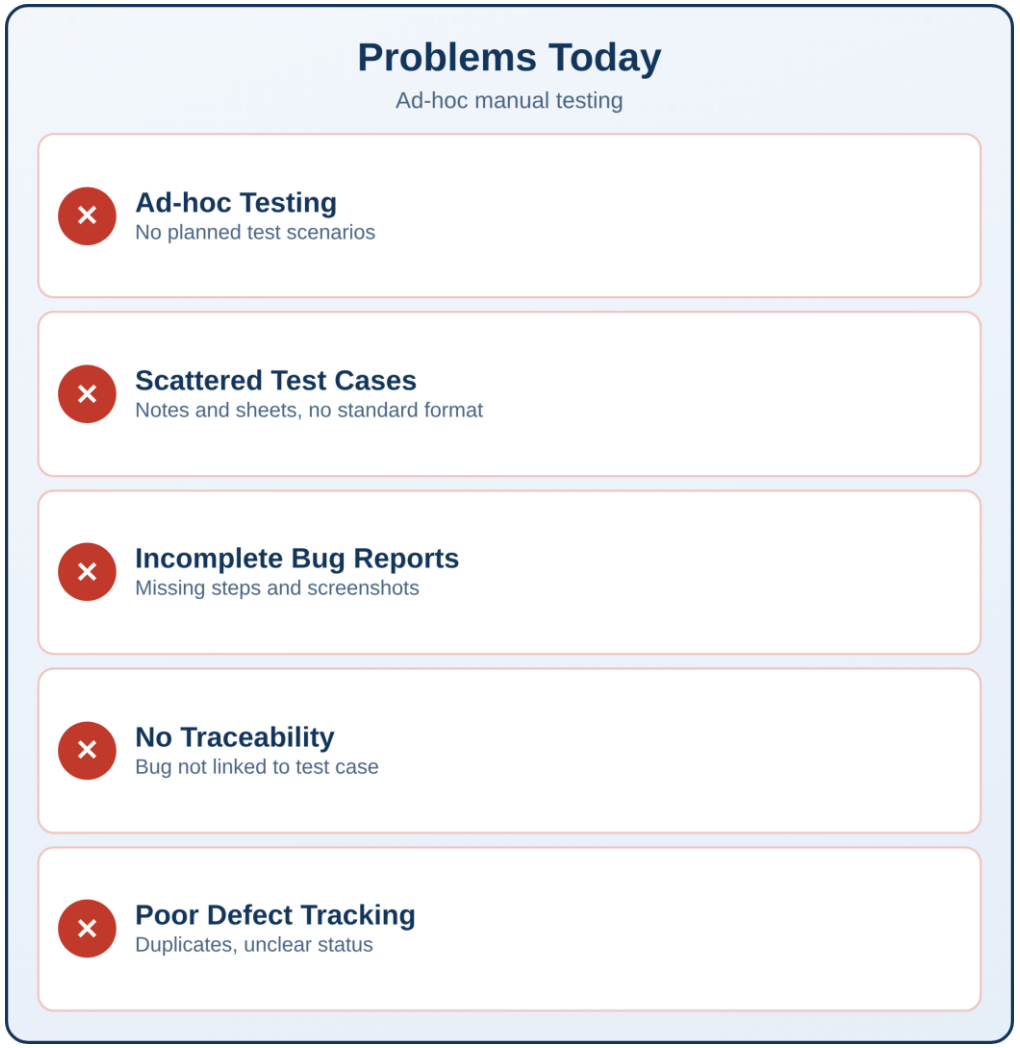


LITERATURE SURVEY

S.No	Title	Authors	Year	Techniques Used	Problem Addressed
1	What Makes a Good Bug Report?	T. Zimmermann, R. Premraj, N. Bettenburg, S. Just, A. Schröter, C. Weiss	2010	Developer survey, CUEZILLA bug report quality scoring	Bug reports often miss steps to reproduce and other details developers need
2	Model Based Testing Approach for Tourism Applications: Use Case Scenarios for Booking Accommodations	E. B. Giritli, Y. S. Can, A. Sen, F. Alagöz	2024	Model-based testing, GraphWalker, Selenium	Reliability of the booking journey: login, hotel search, reservation and payment
3	Hotel Reservation Application Testing at LeGreen Hotel & Suite using Black Box Boundary Value Analysis	A. Yani, D. Setiawan, N. E. Sofian, R. Subagja, T. Desyani	2020	Black box testing, Boundary Value Analysis	Verifying hotel reservation functions; faults found in admin login and print report
4	The Art of Software Testing, 3rd ed.	G. J. Myers, C. Sandler, T. Badgett	2011	Equivalence partitioning, boundary value analysis, cause-effect	Need for systematic design of effective test cases
5	Certified Tester Foundation Level Syllabus v4.0	ISTQB	2023	Test design techniques, defect management	Common terms and practice for test design and defect reporting

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EXISTING SYSTEM

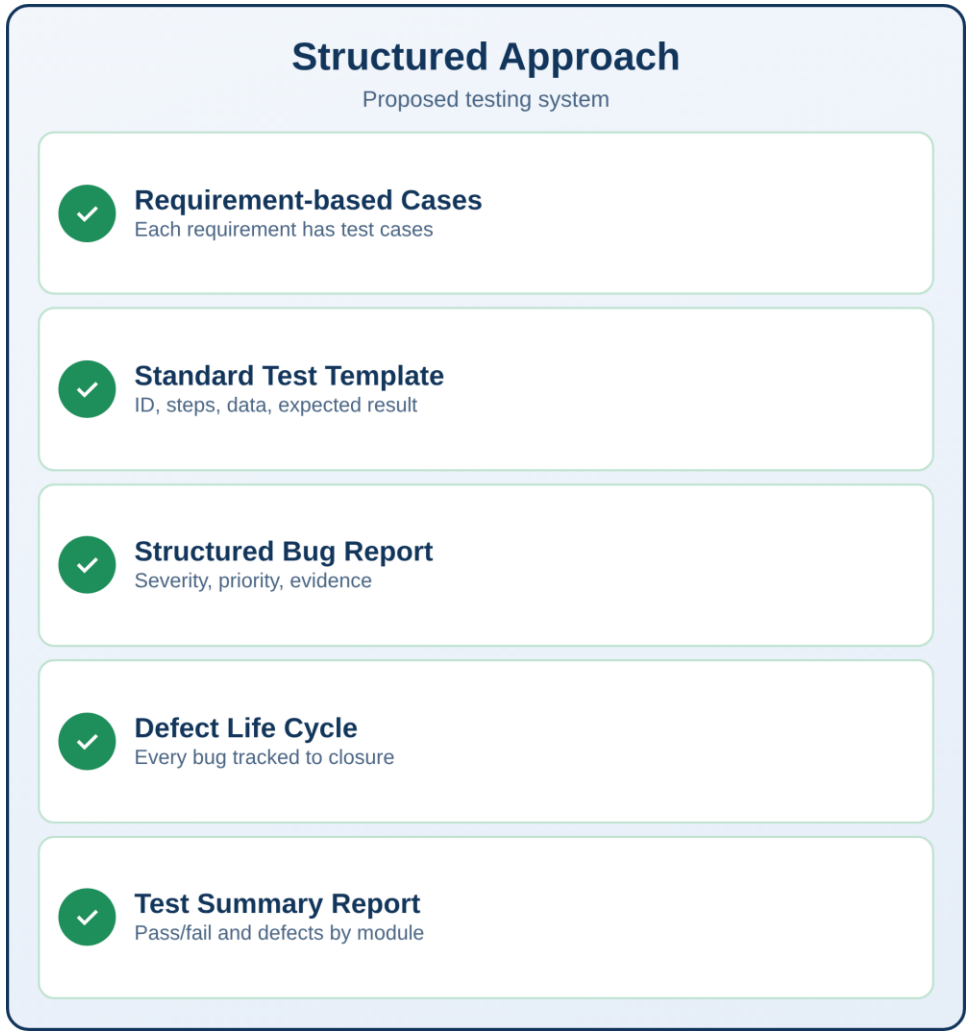


- **Ad-hoc Testing:** Testing is done without a plan, so important scenarios are often missed.
- **Scattered Test Cases:** Test cases are kept in notes or different sheets with no standard format.
- **Incomplete Bug Reports:** Reports often miss steps to reproduce, test data or screenshots, so developers cannot repeat the bug.
- **No Traceability:** Bugs are not linked to the requirement or test case that found them.
- **Poor Defect Tracking:** Duplicate bugs and unclear status make it hard to know what is fixed.

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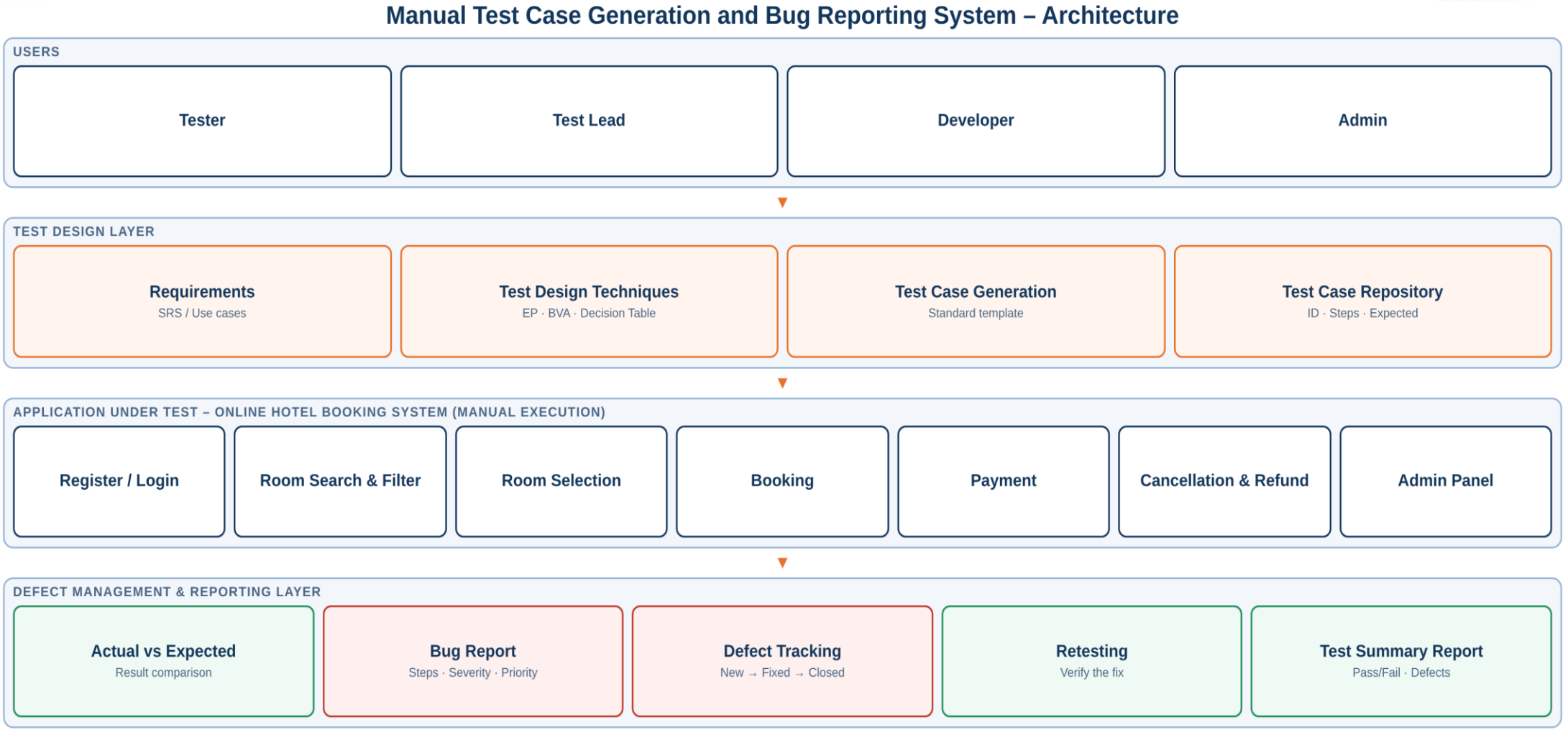
PROPOSED SYSTEM

- ✓ **Requirement-based Test Cases:** Every requirement of the hotel system gets test cases built with EP, BVA and decision tables.
- ✓ **Standard Test Case Template:** Each case has ID, module, precondition, steps, test data, expected result, actual result and status.
- ✓ **Structured Bug Report:** A fixed format with summary, steps, severity, priority, environment and evidence.
- ✓ **Defect Life Cycle:** Each bug moves through New, Assigned, Fixed, Retest and Closed, or Reopened if it fails again.
- ✓ **Test Summary Report:** Shows passed and failed cases and defects by module and severity to support release decisions.



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ARCHITECTURE DIAGRAM



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SAMPLE TEST CASES

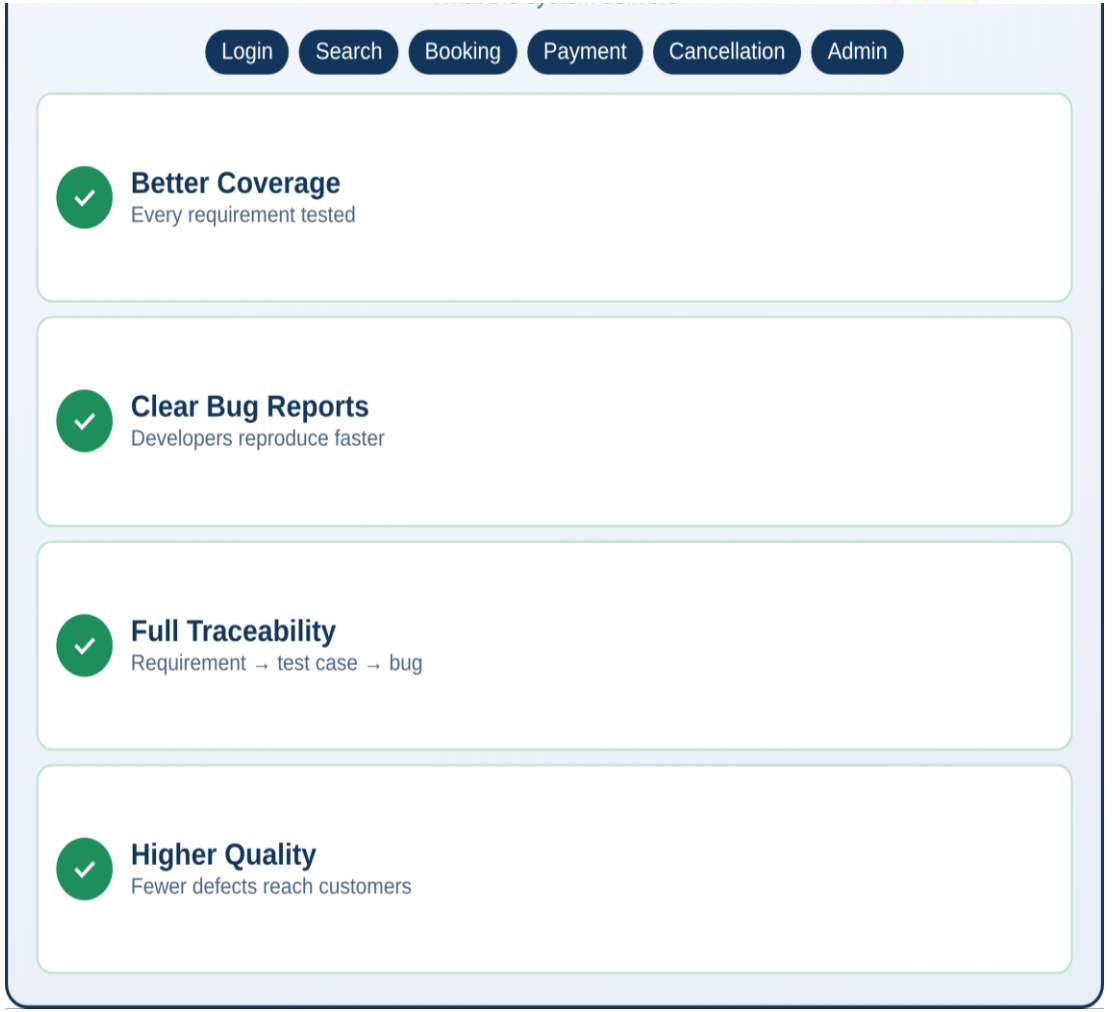
TC ID	Module	Test Scenario	Test Data	Expected Result	Status
TC_01	Login	Login with valid email and password	Email: user@test.com, valid password	User logs in and lands on home page	Pass
TC_02	Search	Search hotels with check-out date before check-in date	Check-in 10-Nov, check-out 08-Nov	Error: check-out must be after check-in	Pass
TC_03	Booking	Book a room for the maximum allowed guests (BVA)	Guests = 4 (max), 5 (above max)	4 accepted; 5 shows error message	Fail
TC_04	Payment	Pay using a card with insufficient balance	Valid card, balance below amount	Payment declined; booking not confirmed	Pass
TC_05	Cancellation	Cancel a confirmed booking before the deadline	Booking ID of confirmed booking	Booking cancelled; refund message shown	Pass

SAMPLE BUG REPORTS

Bug ID	Module	Summary	Steps to Reproduce	Severity / Priority	Status
BUG_01	Booking (TC_03)	System accepts 5 guests when maximum is 4	1. Search a room 2. Select 5 guests 3. Click Book Now	Medium / High	New
BUG_02	Payment	Booking confirmed even though payment failed	1. Choose room 2. Enter declined card 3. Submit	High / High	Assigned
BUG_03	Search	Price filter shows rooms above the selected maximum	1. Open search 2. Set max price 3,000 3. Apply filter	Medium / Medium	Fixed
BUG_04	Home page	Logo overlaps menu on small screens	1. Open site on mobile 2. View home page	Low / Low	Closed

CONCLUSION

This project presents a manual testing system for an online hotel booking website. It designs test cases for login, search, booking, payment and cancellation, executes them by hand, and records every defect in a standard bug report that is tracked to closure. This improves test coverage, helps developers fix bugs faster and raises the quality of the booking system.



REFERNCES

1. T. Zimmermann, R. Premraj, N. Bettenburg, S. Just, A. Schröter, and C. Weiss, “What Makes a Good Bug Report?,” IEEE Transactions on Software Engineering, vol. 36, no. 5, pp. 618–643, 2010.

2. E. B. Giritli, Y. S. Can, A. Sen, and F. Alagöz, “Model Based Testing Approach for Tourism Applications: Use Case Scenarios for Booking Accommodations,” Computer Science & Information Technology (CS & IT), pp. 57–64, 2024. DOI: 10.5121/csit.2024.142006.

3. A. Yani, D. Setiawan, N. E. Sofian, R. Subagja, and T. Desyani, “Pengujian Aplikasi Reservasi Hotel di LeGreen Hotel & Suite dengan Metode Black Box Testing Boundary Value Analysis,” Jurnal Teknologi Sistem Informasi dan Aplikasi, 2020.

4. G. J. Myers, C. Sandler, and T. Badgett, The Art of Software Testing, 3rd ed. Hoboken, NJ, USA: Wiley, 2011.

5. ISTQB, “Certified Tester Foundation Level Syllabus,” International Software Testing Qualifications Board, v4.0, 2023.

THANK YOU